ONLINE RETAIL II

Data Quality and Preparation Report

Python-generated portfolio report

Report generated: 2026-10-29

Purpose: document dataset structure, quality issues, preparation decisions, exceptional transactions, and limitations affecting downstream sales analysis.

Dataset milestone	Validated value
Original source records	1,067,371
Rows removed for identified period overlap	22,523
Rows in analysis dataset	1,044,848
Original columns	8
Standardized prepared columns	10

1. Executive Summary

The Online Retail II workbook contains two source sheets covering December 2009 through December 2011. Preparation standardized the schema, created a line amount, preserved quality flags, investigated missing descriptions and negative amounts, and removed a confirmed overlap between the source periods. The final analysis dataset contains 1,044,848 rows. Questionable records were investigated and documented rather than indiscriminately deleted.

2. Source Data

Source sheet	Rows	Date range
Year 2009-2010	525,461	2009-12-01 07:45 to 2010-12-09 20:01
Year 2010-2011	541,910	2010-12-01 08:26 to 2011-12-09 12:50

Original fields: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, and Country. The standardized dataset uses lowercase names and adds source_period and line_amount.

3. Data Quality Findings

Quality check (full combined dataset)	Rows	Share of 1,067,371
Missing customer ID	243,007	22.77%
Missing description	4,382	0.41%
Cancellation invoice (starts with C)	19,494	1.83%
Negative quantity	22,950	2.15%
Zero unit price	6,202	0.58%
Negative unit price	5	<0.01%
Rows in exact-duplicate groups	23,430	2.20%
Extra exact-duplicate occurrences	12,133	1.14%

These quality counts describe the full combined flagged dataset (1,067,371 rows), before overlap removal. Categories may overlap; cancellation invoices commonly also have negative quantities.

4. Key Investigations

Dataset preparation

The two sheets contain 1,067,371 records. The standardized combined dataset has 10 columns, including source_period and line_amount. Each line amount is quantity multiplied by unit price.

Missing values

Customer IDs are missing in 243,007 rows (22.77%). Such rows may still support product, country, and date analyses but not reliable customer identification. Descriptions are missing in 4,382 rows (0.41%).

Description investigation

Of 4,382 rows with missing descriptions, 4,019 had the same stock code with a non-empty description elsewhere and 363 did not. Across 2,451 affected stock codes, 2,096 had at least one candidate description and 355 had none. Some stock codes have conflicting descriptions, so automatic replacement is unsafe without review.

Negative amounts

In the analysis dataset after overlap removal, 19,169 rows have negative line amounts. Of these, 19,164 have cancellation invoice IDs beginning with C and negative quantities. Five negative-amount rows have negative unit prices; categories overlap and are not mutually exclusive.

Overlapping source periods

The sheets overlap from 2010-12-01 through 2010-12-09. Investigation identified 22,523 matching transaction rows. The analysis dataset excludes these rows from the 2009-2010 source period only, retaining 1,044,848 rows. The original workbook and full flagged dataset are preserved.

Extreme cancellation pair

On 2011-12-09, invoice 581483 records +80,995 units of stock code 23843 at 2.08 per unit (+168,469.60), followed 12 minutes later by cancellation invoice C581484 for -80,995 units and -168,469.60. Customer, product, country, quantity magnitude, and amount magnitude match. Document the pair rather than automatically removing it from every analysis.

Incomplete final month

December 2011 ends on 2011-12-09 12:50 and is incomplete. Do not compare it directly with complete months without qualification or an adjusted comparison window.

5. Preparation Rules and Decisions

Decision	Rule applied / recommended treatment
Preserve raw data	Keep the original workbook unchanged; prepare copies.
Standardize schema	Use consistent lowercase column names and retain source_period.
Line amount	Calculate line_amount = quantity × unit_price; do not automatically call its sum net sales.
Period overlap	Exclude 22,523 matching overlap rows from the 2009-2010 source period in the analysis dataset.
Quality flags	Retain rows in the flagged dataset and add explicit flags instead of silently deleting questionable records.
Missing customer IDs	Retain for product/country/date analyses; treat customer-level results cautiously.
Missing descriptions	Do not automatically replace ambiguous descriptions; review candidate descriptions first.
Cancellations / returns	Use invoice IDs, quantities, and amounts together; distinguish positive, negative, and net measures.
Duplicates	Flag exact duplicates and investigate before deletion because repeated lines may be meaningful.
Extreme values	Document business context before excluding exceptional records from a metric.

6. Limitations and Analytical Risks

• Rows represent transaction lines, not necessarily distinct orders or customers.

• Missing customer IDs limit customer-level analysis and repeat-purchase identification.

• The sum of line_amount is not automatically net sales because cancellations, adjustments, and exceptional records are included.

• Some zero- or negative-price records may represent fees, adjustments, samples, or data-quality issues and require context.

• The source periods overlap and the final month is incomplete; comparisons must account for both.

• The dataset is historical and should not be assumed to represent current retail behavior.

7. Recommended Next Steps

• Define each business metric before filtering records (for example, positive line value, cancellations, or net line value).

• For every KPI or chart, document whether cancellations, zero-price rows, adjustments, and missing customer IDs are included.

• Use comparable complete months for monthly trends and label December 2011 as partial.

• Continue analysis of monthly patterns, countries, products, invoices, and customer behavior where IDs are available.

• Create a data dictionary and README describing the workflow and known limitations.

• Keep reproducible scripts and derived outputs in their project folders; check data-sharing terms before publishing raw data.

8. Project Files Referenced

File	Purpose
02_Processed_Data/online_retail_combined.csv	Standardized combined dataset before overlap removal.
02_Processed_Data/online_retail_flagged.csv	Combined dataset with quality flags.
02_Processed_Data/online_retail_analysis.csv	Analysis dataset after removing identified overlap.
04_Documentation/description_candidates.csv	Candidate descriptions for stock codes with missing descriptions.
03_Results/monthly_sales_overview.csv	Monthly transaction and line-amount overview.
03_Results/monthly_sales_components.csv	Monthly positive, negative, and zero line-amount components.

File	Purpose
03_Results/negative_amount_diagnostics.csv	Diagnostics for negative line amounts.

This report summarizes validated findings recorded during the project workflow. If the data or cleaning rules change, rerun the relevant validation scripts and update this report before publishing it as a portfolio artifact.